Proof Practice

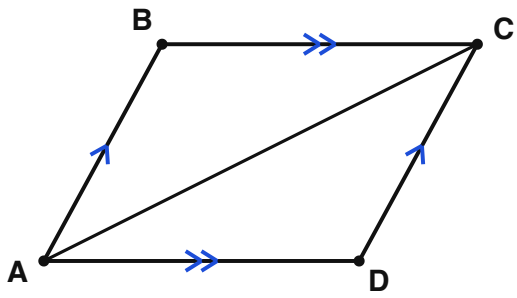
Parallelograms, rhombi, rectangles, squares, kites and converses I diagrams are not drawn to scale

How to attack a proof: (1) Copy the marks from the diagram into Given. (2) Look at what you must Prove and work backward: which theorem or triangle congruence would give that? (3) Fill the table one step at a time, with a reason for every statement. Blue tick marks mean ≅ sides, arrows mean ∥ sides, small squares mean right angles, arcs mean ≅ angles. Stars show difficulty; the harder ones use **converses**.

Theorem bank (from your notes):

- **Right angles congruence:** all right angles are ≅
- **Vertical angles congruence:** vertical angles are ≅
- **Corresponding ∠'s:** ∥ lines, so corresponding ∠'s ≅ (converse: ∠'s ≅, so lines ∥)
- **Alternate interior ∠'s:** ∥ lines, so alternate interior ∠'s ≅
- **Triangle sum:** interior ∠'s of a triangle add to 180°
- **Base ∠'s:** 2 sides of a △ ≅, so the angles opposite are ≅ (converse: 2 ∠'s ≅, so the sides opposite are ≅)
- **Triangle congruence:** SSS, SAS, ASA, AAS, HL
- **CPCTC:** corresponding parts of ≅ triangles are ≅
- **Perpendicular bisector thm:** a point on the perpendicular bisector of a segment is equally far from both endpoints (converse: equally far from both endpoints, so it lies on the perpendicular bisector)
- **Parallelogram opposite sides / opposite ∠'s:** both are ≅ (converses: if both pairs are ≅, it is a parallelogram)
- **Parallelogram consecutive ∠'s:** supplementary
- **Parallelogram diagonals:** bisect each other (perpendicular only in a rhombus or square)
- **Rhombus:** 4 ≅ sides. **Rectangle:** 4 right ∠'s. **Square:** 4 ≅ sides and 4 right ∠'s. **Kite:** 2 pairs of adjacent ≅ sides

1. Diagonal splits a parallelogram ★☆☆



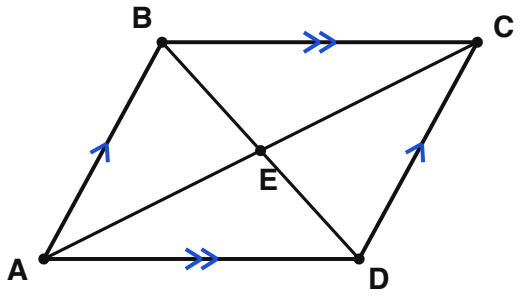
Given: ABCD is a parallelogram with diagonal AC.

Prove: △ABC ≅ △CDA

Tools: Parallelogram opposite sides thm · reflexive property · SSS

	Statements	Reasons
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2. Diagonals of a parallelogram bisect each other ★★☆



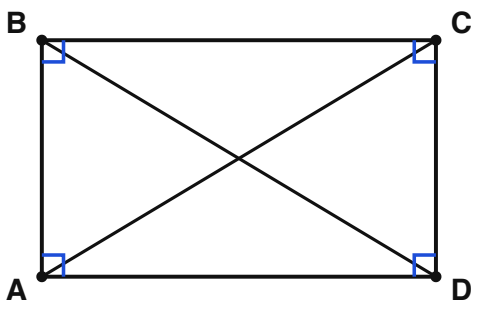
Given: ABCD is a parallelogram. Diagonals AC and BD intersect at E.

Prove: AE ≅ CE and BE ≅ DE

Tools: Alternate interior ∠'s thm · parallelogram opposite sides thm · ASA · CPCTC

	Statements	Reasons
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3. Diagonals of a rectangle are congruent ★★☆



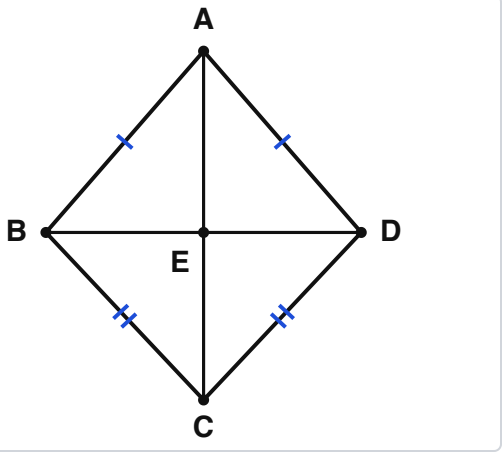
Given: ABCD is a rectangle (four right angles; opposite sides ≅ and ||) with diagonals AC and BD.

Prove: AC ≅ BD

Tools: Right angles congruence thm · parallelogram opposite sides thm · SAS · CPCTC

	Statements	Reasons
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4. A kite's diagonal is a perpendicular bisector (uses a converse) ★★☆



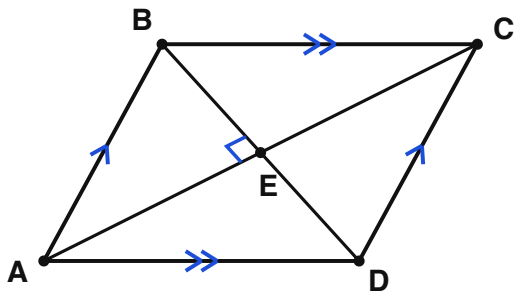
Given: ABCD is a kite with AB ≅ AD and CB ≅ CD. The diagonals meet at E.

Prove: Line AC is the perpendicular bisector of BD.

Tools: Converse of the perpendicular bisector thm · two points determine a line

	Statements	Reasons
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5. Parallelogram with perpendicular diagonals is a rhombus ★★★



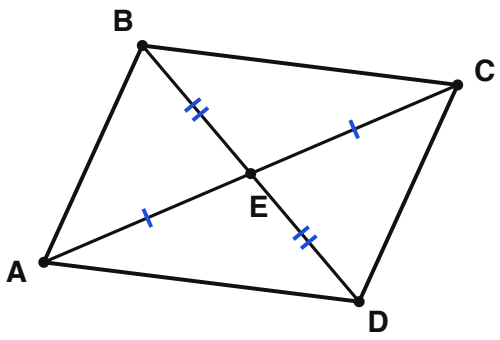
Given: ABCD is a parallelogram. Diagonals AC and BD intersect at E and AC ⊥ BD.

Prove: ABCD is a rhombus.

Tools: Parallelogram diagonals thm · perpendicular bisector thm · parallelogram opposite sides thm · definition of rhombus

	Statements	Reasons
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6. Diagonals that bisect each other make a parallelogram (uses a converse) ★★★



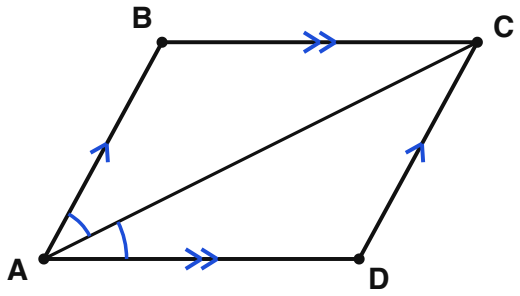
Given: In quadrilateral ABCD, diagonals AC and BD intersect at E with AE ≅ CE and BE ≅ DE.

Prove: ABCD is a parallelogram.

Tools: Vertical angles congruence thm · SAS · CPCTC · converse of the parallelogram opposite sides thm

	Statements	Reasons
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7. Diagonal bisects an angle of a parallelogram: rhombus (uses a converse) ★★★



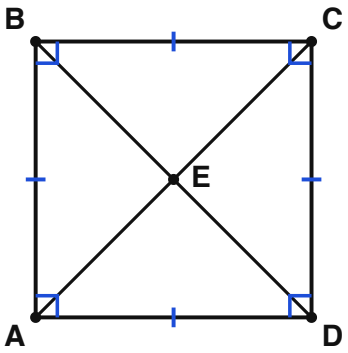
Given: ABCD is a parallelogram. Diagonal AC bisects ∠DAB, so ∠BAC ≅ ∠CAD.

Prove: ABCD is a rhombus.

Tools: Alternate interior ∠'s thm · converse of the base ∠'s thm · parallelogram opposite sides thm

	Statements	Reasons
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8. Challenge: diagonals of a square are congruent and perpendicular ★★★



Given: ABCD is a square with diagonals AC and BD intersecting at E.

Prove: AC ≅ BD and AC ⊥ BD

Tools: Right angles congruence thm · SAS · CPCTC · converse of the perpendicular bisector thm

	Statements	Reasons
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